

BENTLEY C. HARRIS

First-Year Electrical Engineering Student · The University of Queensland

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PROFILE

First-year Electrical Engineering student with two years of formal fabrication and machining training (Certificate II in Engineering) and a growing portfolio of electronics I design and build in my own time: Arduino and ESP32 control systems, circuits laid out in KiCad and hand-soldered, and enclosures designed in Fusion 360 and 3D printed. Most interested in control systems, automation, instrumentation and operational technology. Manual driver's licence and my own car, and available from 21 November 2026 for regional and site-based placements.

EDUCATION

The University of Queensland · St Lucia, QLD 2026 – present

Bachelor of Engineering (Honours) / Master of Engineering, Electrical Engineering. Expected to graduate in 2030.

Courses include ENGG1100 Professional Engineering (team design project), ENGG1001 Programming for Engineers (Python) and ENGG1300 Introduction to Electrical Systems (in progress).

Certificate II in Engineering · Blue Dog Training, completed alongside high school 2024 – 2025

Two-year course assessed through fabricated projects rather than written exams, covering welding, oxy-acetylene cutting, lathe work, sheet metal fabrication, heat treatment, precision measurement and technical drafting.

St James Lutheran College · Hervey Bay, QLD Graduated 2025

- Dux of School 2025, ATAR 93.95.
- Physics 97/100, including 50/50 on internal assessments. Academic Endeavour Award and Optimus Scholaris Award, 2025.
- Represented the school at the 2025 Queensland Year 11/12 Mathematics Competition in Bundaberg.

TECHNICAL SKILLS

Electronics & embedded Arduino and ESP32 microcontrollers, sensor integration (IMU, RTC), PID control, motor drivers (TB6612FNG, L293D), MOSFET switching, buck converters, OLED menu interfaces, RF modules and ESP32 Wi-Fi, battery and supercapacitor monitoring, through-hole soldering on perfboard and stripboard, fault-finding.

Programming Arduino (C/C++) and Python (ENGG1001), both at a working level.

Design & software KiCad schematics and board layouts built onto hand-soldered boards, Fusion 360 CAD for 3D printing, 3D printer operation (Bambu Lab P1S), data logging and spreadsheet analysis.

Fabrication & workshop MIG, stick and spot welding, oxy-acetylene cutting, metal and wood lathe including threading, drill press, band saw, angle grinder, belt sander, heat treating, sheet metal layout and forming.

Measurement & drawing Calipers, scribes and lathe DRO, dimensioned engineering drawings, flat-pattern development, hand-drawn schematics.

ENGINEERING PROJECTS

Full write-ups with photos and video at bentley-harris.github.io

Self-Stabilising Floating House · ENGG1100 team project, team leader 2026

- Led a team building flood-resilient housing that holds itself level and in position on moving water, powered only from the test rig's 12 V, 2 A supply. I designed and built all of the electronics and controls, and designed and made the 3D printed structure and winch system, while my teammates took on most of the theory and the written report.
- Arduino Nano reading an LSM6DS3TR-C IMU through a 50 Hz complementary filter and running a roll-axis PID loop with anti-windup, driving four winch motors through two TB6612FNG drivers. Split motor and logic power rails, a stripboard stack laid out in KiCad, and a handheld operator pod with knobs for live tuning.
- It held the 1 kg shifting load in the test tank, but the tuning was never fully resolved and some oscillation remained. Logging every IMU reading and analysing the overshoot in spreadsheets moved it further than guessing ever did.

Handheld Message Console · personal project, built as a gift 2026

- A 3D printed, Game Boy-style handheld holding 365 messages, one for each day. Arduino Nano, 0.96-inch OLED menu with buttons and buzzer, favourites saved through power-off, a settings menu, and live battery voltage with a low-battery warning.
- The messages would not fit in the Nano's memory as plain text, so I wrote my own compression scheme that stores repeated words and phrases once and rebuilds each message at display time.

Fish Tank Automation System · personal project 2026

- Arduino Nano with a real-time clock and OLED menu that runs the tank light on a schedule, switching the existing LED channels through MOSFETs, and drives a 3D printed screw feeder that has run unattended for more than 30 consecutive days.
- Designed around what could go wrong: a screw feeder that jams keeps the food trapped instead of dumping it, and the firmware limits feeding to two feeds in 24 hours and one per hour.

Wireless and ESP32 Projects · personal projects

2026

- Built a Wi-Fi access point on an ESP32-C3 as a personal test network, with a 0.96-inch I2C OLED showing the network details, connected clients and uptime, and one button to cycle between screens.
- Built and tested an RF remote control for the fish tank controller. It worked, but I cut it from the final design to keep an unattended system as simple and reliable as possible.
- Next step: moving the fish tank controller from an Arduino Nano to an ESP32-C3 for its smaller size and onboard Wi-Fi, so it can connect to my phone.

Supercapacitor Bank · personal project

2025

- My first electronics project taken all the way to a finished unit: controlled charging, per-cell voltage and charge rate on an OLED, overcharge warnings, status LEDs, arming and key switches, and controlled discharge through clamp terminals I machined from scrap aluminium.
- The cells did not charge evenly, and a pack-level reading hid it completely. Since then, almost everything I build reports what it is doing internally.

Workshop Fabrication · Certificate II projects

2024 – 2025

- Worked from engineering drawings, some of them my own. Built a G-clamp from raw stock to a formal drawing pack with a 14-item parts list, including a lathe-machined screw and pad and a plug-welded laminated frame.
- Designed and built a folding camping cooktop from my own dimensioned drawings, with a welded frame and lathe-turned pivot bushes. Also a sheet metal toolbox in 0.5 mm Galvabond and a stepped plumb bob turned to a supplied drawing.

Smaller builds

An LED heart being rebuilt with two 555 timers (PWM dimming, adjustable flash rate and a resistor for every LED), and CAD-designed adapters for a bench solder fume extractor.

LICENCES & AVAILABILITY

- Queensland driver's licence (manual), and my own car.
- Responsible Service of Alcohol (RSA).
- Available for regional, remote and site-based placements for the full duration, from 21 November 2026.

WORK EXPERIENCE

Bartender and Waiter · Tres Salsas, Hervey Bay

Sep 2024 – Feb 2026

- Worked in a small team through busy services such as New Year's Eve, coordinating the bar, kitchen and floor to keep things moving.
- Served alcohol under RSA requirements, counted and secured the till, and dealt with complaints and problems on the spot.
- Helped make the team's TikTok videos, which reached tens of thousands of views and brought customers in.

Night Fill Team Member · Fresh & Save, Hervey Bay

Nov 2023 – Nov 2024

- Restocked the store to a fixed nightly deadline as part of a small crew, working to set procedures with minimal supervision, alongside full-time study and a second casual job.

Casual Grounds Assistant · Hervey Bay Garden Centre

2+ years

- Looked after plant stock across the site on a recurring care schedule, working on my own to a consistent standard.

Work Experience Placement · Health Department, Fraser Coast Regional Council

2023

- 45 hours with a local government health department, seeing how regulatory inspection and compliance work in practice. Reference available.

CLUBS, LEADERSHIP & VOLUNTEERING

- Member of EBESS and UQ MARS (2026). Their PCB design workshops and CAD advice noticeably improved my group's ENGG1100 project.
- Senior Sports Leader, St James Lutheran College (2024): organised and ran house sporting events.
- Formal Planning Committee (2025): helped allocate the budget, including a group decision to change venues that freed up money for other parts of the event.
- Debating Club (2025, 8 months) and regular volunteer at school events, including GALA days.
- Volunteer marker, UQ QAMT Mathematics Competition (2025). Selected for the UQ InspireU Year 12 Camp (2025).